

Farah Mohamed

mohameffarah1@gmail.com | [LinkedIn](#) | [Github](#) | [Portfolio](#)

EDUCATION

Université de Montréal

Computer Science (B.Sc.)

Montréal, QC

2022 – 2025

PROJECTS

Finance Management App | Python, FastAPI, React, TypeScript, HTML, CSS, Gemini AI

[Github](#) | [Site](#)

- Developed a RESTful API backend using Python and FastAPI to automate bank statement parsing.
- Integrated Google Gemini AI API with custom prompt engineering to automatically categorize transactions.
- Developed a responsive dashboard using React, TypeScript, HTML and Tailwind CSS that displays spending summaries and lets users search and edit categories, providing real-time financial insights.
- Containerized the backend with Docker for deployment on Railway and hosted the frontend on Vercel.
- Implemented client-side caching to improve user experience with instant data retrieval.

AI Inbox Firewall | Python, Django, React, Solace Agent Mesh, PostgreSQL, Redis, Docker

[Github](#) | [Demo](#)

- Developed an intelligent email inbox featuring a 6 specialized AI agents architecture utilizing Solace Agent Mesh.
- Built a user interface using React to monitor asynchronous agent workflows and display live inbox status updates.
- Orchestrated the full-stack environment using Docker Compose to containerize Django REST API, React, Solace PubSub+, PostgreSQL, and Redis.

Collaborative Whiteboard | Next.js, React.js, Typescript, HTML, CSS

[Github](#) | [Site](#)

- Developed a Full Stack real-time collaborative whiteboard application allowing multiple users to draw and interact simultaneously.
- Implemented secure authentication with Clerk and real-time synchronization with Liveblocks.
- Utilized Convex as a reactive database and Next.js for a modern and high-performance web architecture.

Real-Time Audio Translation | Next.js, React.js, Typescript, HTML, CSS, REST API

[Github](#) | [Site](#)

- Automated multilingual communication application with real-time processing.
- Resilient architecture with multiple APIs for high availability.
- Collaboration using GitHub and automated deployment on GitHub Pages.

Compiler Host Language Integration | Scheme, BASIC, x86

[Github](#)

- Four-month project conducted for a Programming Languages and Compilation course (IFT3065).
- Utilized Scheme and x86 assembly to implement a parser, lexer, and target code generator.
- Collaborated via GitHub in a team repository, resulting in the final integration of the host language.

EXPERIENCE

Security Analyst

July 2024 – June 2025

Canada Revenue Agency (CRA)

Ottawa, ON

- Analyzed security risks of software and created accounts to approve or deny requests for additional permissions.
- Consulted directly with requesters to clarify security requirements and gather critical information for rigorous risk assessments.

Tutor

October 2023 – June 2025

MonsieurMath

Montréal, QC

- Personalized guidance for more than 20 students, leading to academic success with an average score above 80%.
- Optimized remote learning through the collaborative use of Teams and Miro for the visualization of complex concepts.
- Designed iterative learning plans based on the specific objectives of each student.
- Analyzed and managed pedagogical data via Excel to automate progress tracking.

TECHNICAL SKILLS

Programming Languages: Python, Java, SQL, TypeScript, JavaScript, Haskell, C, C++, Scheme, Swift, Gambit, BASIC

Frameworks: Next.js, React, NumPy, PyTorch, Node.js, Three.js

Developer Tools: Git, VS Code, Visual Studio, Docker, Microsoft 365 suite, Figma, Miro

Languages: French (Native), English (Fluent), Spanish (Conversational)